SUMMER INTERNSHIP PROJECT

Women's Safety System

A mobile-web application designed to provide rapid emergency response, live location sharing, and one-touch SOS activation for women in distress.

Prepared By

Nandana Nambiar

Project Type

Summer Internship — Computer Science & Engineering

01 — INTRODUCTION

Abstract & Introduction

Women's safety remains a critical concern across the globe. Despite advancements in technology, response times during emergencies are often too slow to prevent harm. The **Women's Safety System** is a lightweight web-based application engineered to bridge this gap by offering an immediate, one-touch mechanism for alerting trusted contacts.

This project leverages widely accessible web technologies to ensure availability across devices without the need for a dedicated mobile application download. By combining real-time location sharing, emergency contact management, and an intuitive user interface, the system empowers users to act decisively in high-stress situations.

Core Mission

Provide a reliable, rapid-response safety tool accessible from any internet-connected device.

Approach

A web app built with open standards — no installation required, minimal learning curve.

02 — SCOPE

Problem Statement & Objectives

The Problem

Existing emergency response mechanisms often rely on manual dialling, delayed location sharing, or fragmented third-party applications. In a crisis, every second counts, and fumbling through apps or typing addresses can cost critical time.

There is a clear need for a single, unified platform that allows a user to send their live location and a distress signal to pre-configured contacts instantly — without unnecessary navigation or authentication steps.

1 One-Touch SOS

Enable instant alert dispatch with a single button press.

2 Real-Time Location

Share live GPS coordinates with designated emergency contacts.

3 Contact Management

Allow users to register, add, and edit a prioritised list of emergency contacts.

4 Cross-Platform Access

Ensure the application functions on both mobile and desktop browsers.

03 — COMPARISON

Existing System vs Proposed System

Parameter	Existing System	Proposed System
Response Mechanism	Manual phone calls or SMS to each contact	Automated simultaneous alert to all saved contacts
Location Sharing	Separate app required for live GPS sharing	Built-in live location sharing via Google Maps API
Accessibility	Heavy native apps; platform-dependent	Browser-based; accessible on any device
User Experience	Cluttered UI; too many steps to trigger an alert	Minimalist, one-screen SOS activation

The proposed system eliminates friction during emergencies. By unifying location sharing, contact management, and alerting into a single progressive web interface, the application significantly reduces the time between a distress event and the notification of help. The architecture ensures that no additional installation is necessary — a critical advantage in situations where every second matters.

04 — CAPABILITIES

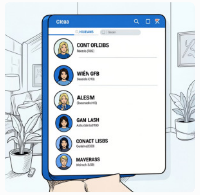
Key Features

SOS Button

A prominent, always-accessible button that instantly sends a distress signal to all saved emergency contacts.

Live Location Sharing

Real-time GPS tracking powered by the Google Maps API, ensuring contacts always know the user's precise location.



Emergency Contacts

Users can add, edit, and prioritise a list of trusted individuals who will receive alerts when SOS is triggered.

Login & Registration

Secure Firebase-based authentication protects user data and ensures only authorised access.

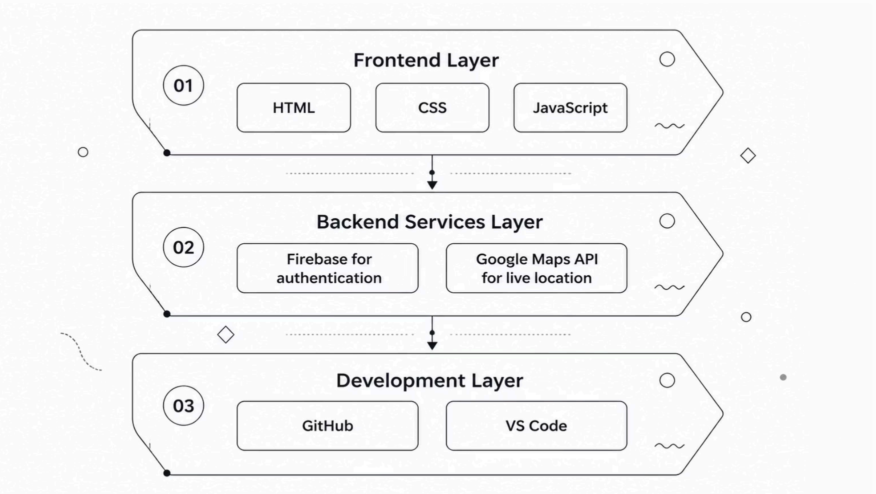
User-Friendly Interface

A minimalist, distraction-free design prioritising accessibility and ease of use under stress.

Technologies Used

HTML5	Semantic structuring of application pages
CSS3	Responsive styling, flexbox layout, and theming
JavaScript	Client-side logic, SOS event handling, and DOM manipulation
Firebase	User authentication and real-time database
Google Maps API	Live location sharing and mapping
GitHub	Version control and repository hosting
VS Code	Primary development IDE

Below is a simplified stack diagram showing how each technology contributes to the overall system. The front-end layer handles presentation and interaction, while Google Maps and Firebase provide essential geolocation and backend services.

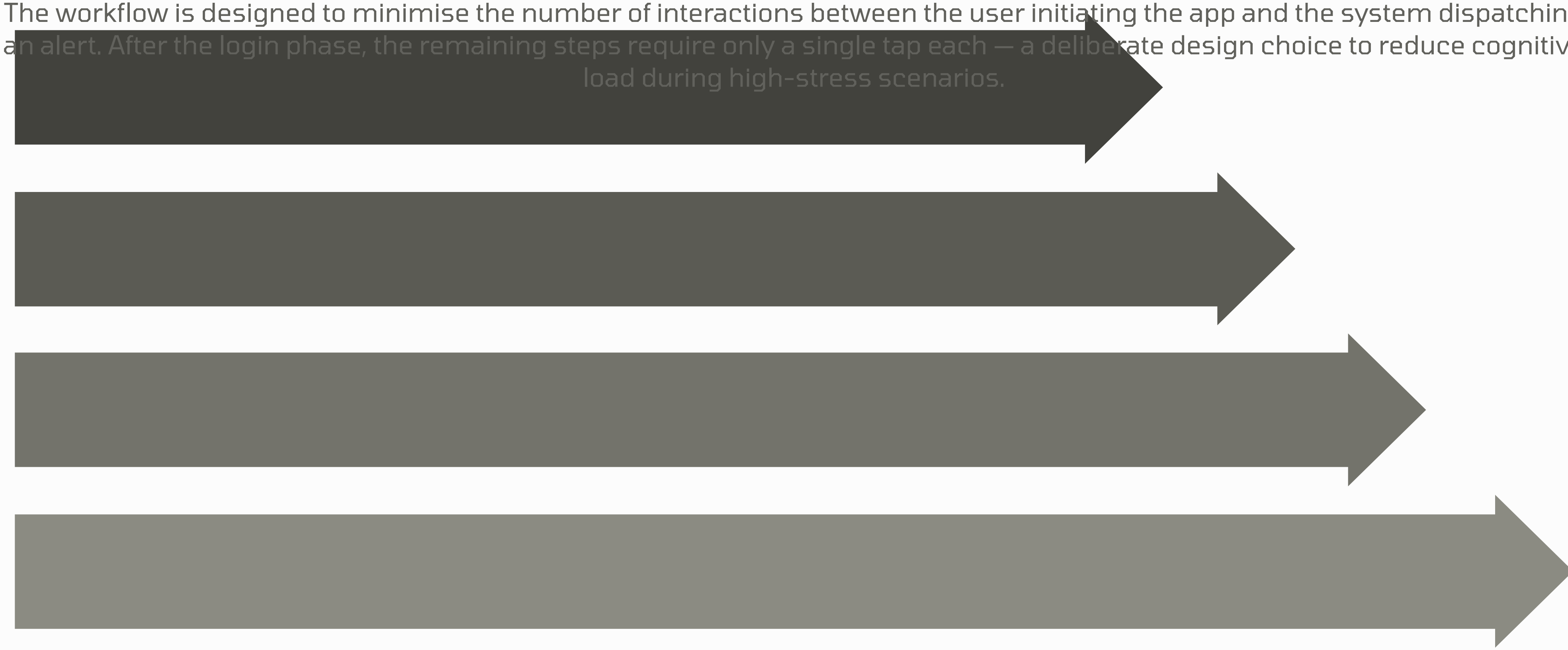


06 — WORKFLOW

System Workflow

The following flowchart illustrates the end-to-end user journey — from initial authentication through to the delivery of emergency assistance. Each node represents a discrete application state or system action.

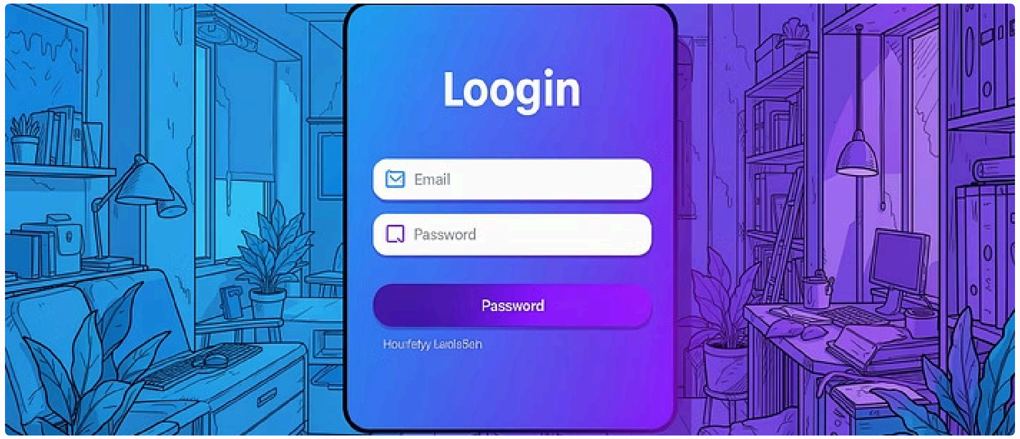
The workflow is designed to minimise the number of interactions between the user initiating the app and the system dispatching an alert. After the login phase, the remaining steps require only a single tap each — a deliberate design choice to reduce cognitive load during high-stress scenarios.

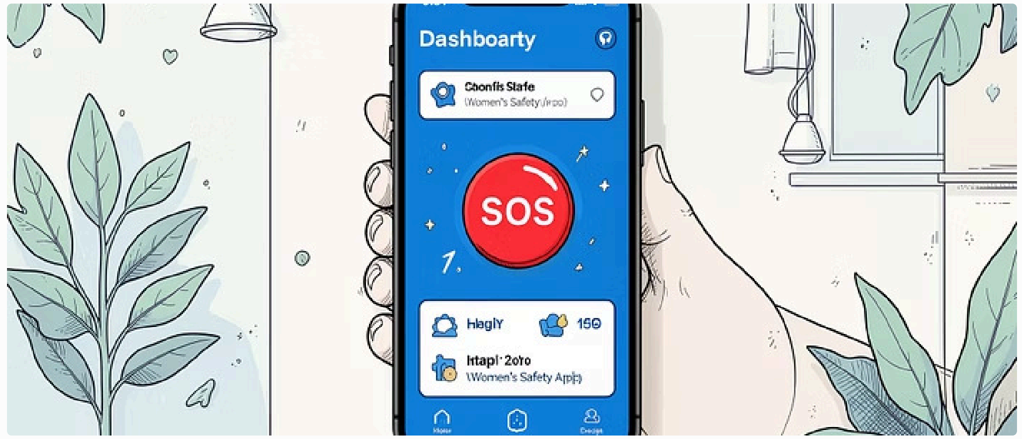


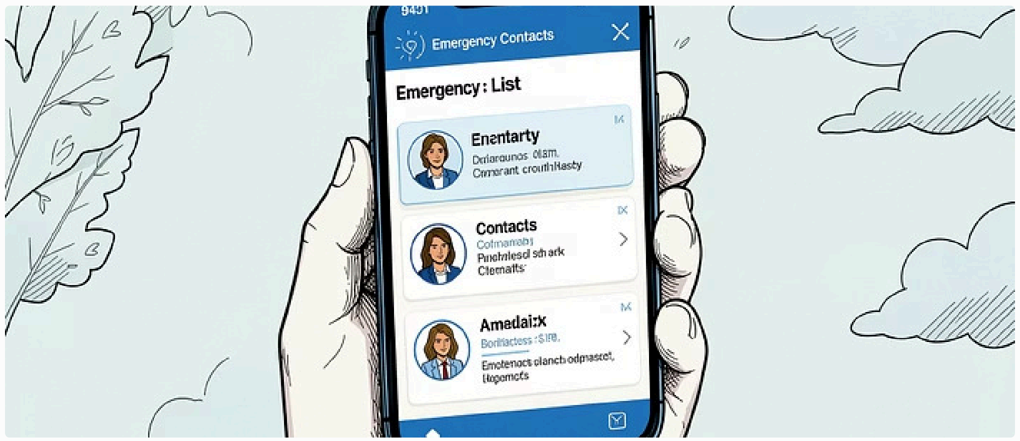
07 — IMPLEMENTATION

Implementation & Screenshots

The application was developed iteratively. The login module and Firebase authentication were implemented first, followed by the home dashboard, contact management screens, and finally the SOS logic integrated with the Google Maps location service. Below are key screenshots from the application.







Screenshots shown are illustrative representations of the application's interface.

Results, Challenges & Future Scope

1

Project Outcomes

The final build successfully demonstrated one-touch SOS activation, live GPS sharing, and Firebase-managed user sessions. The interface rendered responsively across mobile and desktop browsers.

2

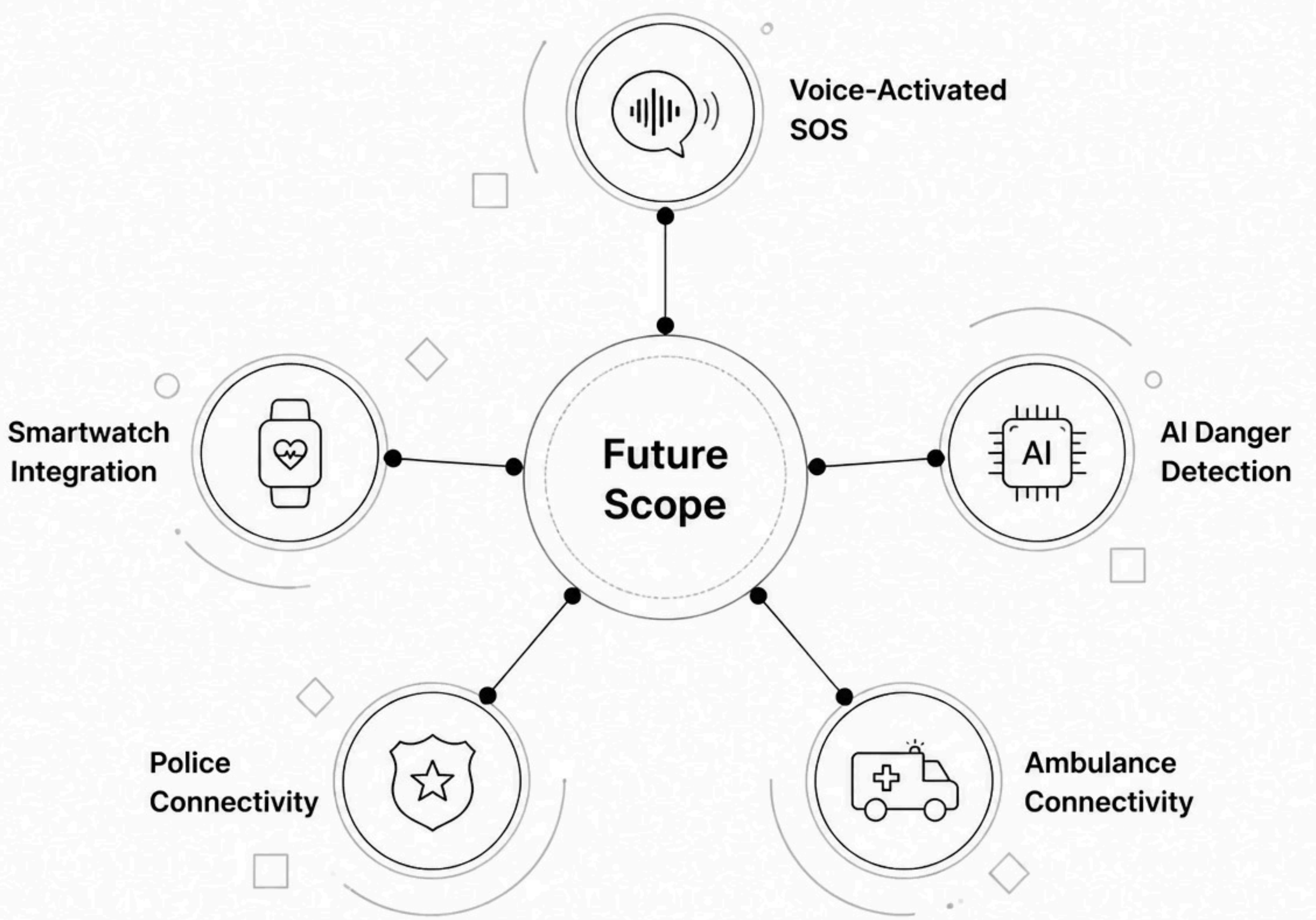
Development Challenges

Key hurdles included fine-tuning the Google Maps API refresh rate without draining battery, ensuring location accuracy indoors, and handling unauthorised database reads via Firebase security rules.

3

Future Enhancements

Planned features include voice-activated SOS, AI-based anomaly detection using accelerometer data, native smartwatch integration, and direct automated connectivity to police and ambulance dispatch services.



09 — CLOSING

Conclusion & References

Conclusion

The Women's Safety System fulfils its primary objective of providing a dependable, rapid-response platform for emergency alerting. By leveraging standard web technologies — HTML, CSS, JavaScript, Firebase, and the Google Maps API — the project demonstrates that accessible, impactful safety solutions do not require heavy native applications or expensive infrastructure.

The internship provided valuable experience in full-stack web development, real-time database integration, and user-centred design constrained by real-world usability requirements. Future iterations incorporating AI detection and smartwatch support will further strengthen the platform's reliability and reach.

References

- MDN Web Docs — HTML5 Specification
- MDN Web Docs — CSS Reference Guide
- MDN Web Docs — JavaScript Documentation
- Firebase Authentication & Realtime Database — Official Docs
- Google Maps JavaScript API — Developers Guide
- GitHub Version Control System — User Guide